

Supplementary Material

8 Notations and mathematical definitions

In this section, we introduce some symbol definitions and the preliminary knowledge, and supplement the specific definitions of related concepts mentioned in Section 2.

8.1 Definitions of norm

We first introduce the definitions of norm. Given a tensor $\mathcal{M} \in \mathbb{R}^{\prod_{i=1}^N I_i}$, let $m_{i_1 i_2 \dots i_N}$ denote the $(i_1, i_2, \dots, i_N)$ -th component of $\mathcal{M}$, then the Frobenius norm of $\mathcal{M}$ is defined as

$$\|\mathcal{M}\|_F := \sqrt{\sum_{i_N=1}^{I_N} \dots \sum_{i_1=1}^{I_1} m_{i_1 i_2 \dots i_N}^2}.$$

8.2 Mathematical definitions for nonconvex analysis

Next, we recall several basic concepts for nonconvex analysis [2,3].

Definition 7. Let S be a set, the derived set S' is defined as the set of overall cluster points of S .

Proposition 6. A bounded closed set in finite-dimensional space is a compact set.

Definition 8. A function $g : \mathbb{R}^n \rightarrow (-\infty, +\infty]$ is said to be proper if $\text{dom } g \neq \emptyset$, where $\text{dom } g = \{x \in \mathbb{R} : g(x) < \infty\}$.

Definition 9. If $\forall x_k \in \text{dom } f$, we have

$$\lim_{k \rightarrow \infty} x_k = x, \quad f(x) \leq \liminf_{k \rightarrow \infty} f(x_k),$$

then f is said to be lower semicontinuous on $\text{dom } f$.

Definition 10. Let f be a proper lower semicontinuous function, the Fréchet subdifferential of f at x , written $\hat{\partial}f(x)$, is the set of all vectors u which satisfy

$$\lim_{y \neq x, y \rightarrow x} \frac{f(y) - f(x) - \langle u, y - x \rangle}{\|y - x\|} \geq 0,$$

when $x \notin \text{dom } f$, then set $\hat{\partial}f(x) = \emptyset$.

Proposition 7. If f is a proper lower semicontinuous function and has a local minimum at x^* , then $0 \in \hat{\partial}f(x^*)$.

Proposition 8. Let f be a proper lower semicontinuous function, and g be a continuously differential function, then we have $\forall x \in \text{dom } f$, $\partial(f + g)(x) = \hat{\partial}f(x) + \nabla g(x)$.

Proposition 9. Set $f: \mathbb{R}^n \rightarrow \mathbb{R}$, if $\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|$, then $f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2}\|y - x\|^2$, where L is the Lipschitz constant of ∇f .

Definition 11. Desingularization function: we denote the desingularization function $\phi: [0, \eta) \rightarrow \mathbb{R}_+$ which satisfies these conditions: (i) $\phi(0) = 0$. (ii) ϕ is continuously differentiable on $[0, \eta)$, and ϕ is a concave function from in $(0, \eta)$. (iii) $\phi(s)' > 0$ ($\forall s \in (0, \eta)$).

Definition 12. Let f be a proper lower semicontinuous function, f is said to have the KL property on $\bar{u} \in \text{dom}(\partial f)$, if there exists $\eta \in (0, +\infty]$, and U is the neighborhood of $\bar{u}$, any u in U that satisfies the condition $f(\bar{u}) < f(u) < f(\bar{u}) + \eta$ has the following inequalities established.

$$\phi'(f(u) - f(\bar{u}))\text{dist}(0, \partial f(u)) > 1,$$

where ϕ satisfy (Definition 11). If f has the KL property at each point of $\text{dom } \partial f$, then f is a KL function.

Proposition 10. (Uniformized KL property) Let Ω be a compact set and $f: \mathbb{R}^n \rightarrow \mathbb{R} \cup \{\infty\}$ be a proper and lower semicontinuous function. Assume that f is constant on Ω and is a KL function. Then, there exist $\epsilon > 0$, $\eta > 0$ such that $\forall x \in \Omega$ and all x in the intersection $\{x \in \mathbb{R}^n : \text{dist}(x, \Omega) < \epsilon\} \cap \{f(\bar{x}) < f(x) < f(\bar{x}) + \eta\}$ one has

$$\phi'(f(x) - f(\bar{x}))\text{dist}(0, \partial f(x)) > 1.$$

9 Convergence analysis

In this section, we prove all the theorems in Section 5.

9.1 Proof of Theorem 1

Proof. (i) If $J(A_{(N)}^{k+1}) \leq J(A_{(N)}^k) - \rho_t \|A_{(N)}^{k+1} - Y_{(N)}^k\|_F^2$, then the Theorem 1(i) is true. If not, since Eq. (7), we know that $H(A_{(N)})$ satisfies Proposition 9. Thus, we have

$$\begin{aligned} H(A_1^{k+1}, \{A_i^k\}_{i=2}^N) &\leq \langle \nabla_{A_1} H(A_{(N)}^k), A_1^{k+1} - A_1^k \rangle \\ &\quad + \frac{L_{\nabla_{A_1} H}}{2} \|A_1^{k+1} - A_1^k\|_F^2 + H(A_{(N)}^k). \end{aligned} \quad (8)$$

From Eq. (9), we can also obtain

$$\begin{aligned} F_1(A_1^k) &\geq F_1(A_1^{k+1}) + \frac{1}{2\sigma_1^k} \|A_1^{k+1} - A_1^k\|_F^2 \\ &\quad + \langle \nabla_{A_1} H(A_{(N)}^k), A_1^{k+1} - A_1^k \rangle. \end{aligned} \quad (9)$$

Then sum of the Eq. (8) and Eq. (9), we have

$$\begin{aligned} H(A_1^{k+1}, \{A_i^k\}_{i=2}^N) + F_1(A_1^{k+1}) &\leq H(A_{(N)}^k) + F_1(A_1^k) \\ &\quad - \rho \|A_1^{k+1} - A_1^k\|_F^2, \end{aligned} \quad (10)$$

where $\rho = \min(\frac{1}{2\sigma_1^k} - \frac{L\nabla_{A_1^k} H}{2})$. Assuming Theorem 1(i) holds when $n = N - 1$, i.e.,

$$H(A_{(N-1)}^{k+1}, A_N^k) + \sum_{i=1}^{N-1} F_i(A_i^{k+1}) \leq H(A_{(N)}^k) + \sum_{i=1}^{N-1} F_i(A_i^k) - \rho \|A_{(N-1)}^{k+1} - A_{(N-1)}^k\|_F^2, \quad (11)$$

where $\rho = \min(\{\frac{1}{2\sigma_i^k} - \frac{L\nabla_{A_i^k} H}{2}\}_{i=1}^{N-1})$. Similarly, from Eq. (9), we infer

$$F_N(A_N^k) \geq F_N(A_N^{k+1}) + \frac{1}{2\sigma_N^k} \|A_N^{k+1} - A_N^k\|_F^2 + \langle \nabla_{A_N} H(A_{(N-1)}^{k+1}, A_N^k), A_N^{k+1} - A_N^k \rangle. \quad (12)$$

From Eq. (7) and Proposition 9, we also have

$$H(A_{(N)}^{k+1}) \leq H(A_{(N-1)}^{k+1}, A_N^k) + \langle \nabla_{A_N} H(A_{(N-1)}^{k+1}, A_N^k), A_N^{k+1} - A_N^k \rangle + \frac{L\nabla_{A_N^k} H}{2} \|A_N^{k+1} - A_N^k\|_F^2. \quad (13)$$

Then sum of the Eq. (11), Eq. (12) and Eq. (13), we have

$$J(A_{(N)}^{k+1}) \leq J(A_{(N)}^k) - \rho \|z^{k+1} - c^k\|_F^2, \quad (14)$$

where $\rho = \min(\{\frac{1}{2\sigma_i^k} - \frac{L\nabla_{A_i^k} H}{2}\}_{i=1}^N)$. This shows the Theorem 1(i) is true.

(ii) We know that $\frac{\alpha}{2} \text{Tr}((A_N)^T L A_N)$ in Eq. (4) is nonnegative. Hence, Eq. (4) is nonnegative. From Eq. (14), we have

$$\rho \|z^{k+1} - c^k\|_F^2 \leq J(A_{(N)}^k) - J(A_{(N)}^{k+1}).$$

Sum of both side, since Eq. (4) is nonnegative, thus we have

$$\begin{aligned} \rho \sum_{k=1}^{\infty} \|z^{k+1} - c^k\|_F^2 &\leq \sum_{k=1}^{\infty} (J(A_{(N)}^k) - J(A_{(N)}^{k+1})) \\ &= J(A_{(N)}^1) - \inf J \\ &< \infty. \end{aligned}$$

It follows that

$$\lim_{k \rightarrow \infty} \|z^{k+1} - c^k\|_F^2 = 0.$$

□

9.2 Lemma: subgradient lower bound property of Algorithm 1

Before continuing with the proof of the theorems of Section 5, we prove the following lemma.

Lemma 2. *If we define*

$$\begin{aligned} f_i^{k+1} &= \nabla_{A_i} H(\{A_j^{k+1}\}_{j=1}^i, \{A_j^k\}_{j=i+1}^N) + \frac{1}{\sigma_i^k} (Y_i^k - A_i^{k+1}) \\ &\quad - \nabla_{A_i} H(\{A_j^{k+1}\}_{j=1}^{i-1}, Y_i^k, \{A_j^k\}_{j=i+1}^N). \end{aligned} \quad (15)$$

Then we have $\|\{f_i^{k+1}\}_{i=1}^N\|_F \leq \rho_b \|z^{k+1} - c^k\|_F$, where ρ_b is a positive constant.

Proof. By Proposition 7 and Proposition 8, Eq. (9) follows that

$$\begin{aligned} 0 &\in \partial_{A_i} F_i(A_i^{k+1}) - \frac{1}{\sigma_i^k} (Y_i^k - A_i^{k+1}) \\ &\quad + \nabla_{A_i} H(\{A_j^{k+1}\}_{j=1}^{i-1}, Y_i^k, \{A_j^k\}_{j=i+1}^N). \end{aligned} \quad (16)$$

Thus we also have

$$\begin{aligned} f_i^{k+1} &= \nabla_{A_i} H(\{A_j^{k+1}\}_{j=1}^i, \{A_j^k\}_{j=i+1}^N) - \frac{1}{\sigma_i^k} (A_i^{k+1} - Y_i^k) \\ &\quad - \nabla_{A_i} H(\{A_j^{k+1}\}_{j=1}^{i-1}, Y_i^k, \{A_j^k\}_{j=i+1}^N) \\ &\in \nabla_{A_i} H(\{A_j^{k+1}\}_{j=1}^{i-1}, A_i^{k+1}, \{A_j^k\}_{j=i+1}^N) \\ &\quad + \partial_{A_i} \left(\sum_{j=1}^i F_j(A_j^{k+1}) + \sum_{j=i+1}^N F_j(A_j^k) \right) \\ &= \partial_{A_i} J(\{A_j^{k+1}\}_{j=1}^{i-1}, A_i^{k+1}, \{A_j^k\}_{j=i+1}^N). \end{aligned} \quad (17)$$

Since Eq. (7), we have

$$\begin{aligned} \|f_i^{k+1}\|_F &= \|\nabla_{A_i} H(\{A_j^{k+1}\}_{j=1}^i, \{A_j^k\}_{j=i+1}^N) + \frac{1}{\sigma_i^k} (Y_i^k - A_i^{k+1}) \\ &\quad - \nabla_{A_i} H(\{A_j^{k+1}\}_{j=1}^{i-1}, Y_i^k, \{A_j^k\}_{j=i+1}^N)\|_F, \\ &\leq \|\nabla_{A_i} H(\{A_j^{k+1}\}_{j=1}^i, \{A_j^k\}_{j=i+1}^N) \\ &\quad - \nabla_{A_i} H(\{A_j^{k+1}\}_{j=1}^{i-1}, Y_i^k, \{A_j^k\}_{j=i+1}^N)\|_F \\ &\quad + \left\| \frac{1}{\sigma_i^k} (A_i^{k+1} - Y_i^k) \right\|_F \\ &\leq (L_{\nabla_{A_i^k} H} + \frac{1}{\sigma_i^k}) \|A_i^{k+1} - Y_i^k\|_F. \end{aligned} \quad (18)$$

Thus we infer

$$\|\{f_i^{k+1}\}_{i=1}^N\|_F \leq \rho_b \|z^{k+1} - c^k\|_F, \quad (19)$$

where $\rho_b = \max(\{\frac{1}{\sigma_i^k} + L_{\nabla_{A_i^k} H}\}_{i=1}^N)$. $\square$

9.3 Proof of Theorem 2

Proof. (i) Observe that z' can be viewed as an intersection of compact sets

$$z' = \bigcap_{q \in \mathbb{N}} \overline{\bigcup_{k \geq q} \{z^k\}}.$$

From proposition 6, z' is compact.

(ii) $\forall \bar{z} \in z'$, there exists a subsequence z^{k_j} such that

$$\lim_{j \rightarrow \infty} z^{k_j} = \bar{z}.$$

Let

$$F(z^{k_j}) = \sum_{i=1}^N F_i(A_i^{k_j}). \quad (20)$$

Since F_i is lower semicontinuous [3], from Definition 9 and Eq. (20), we obtain that

$$\liminf_{j \rightarrow \infty} F(z^{k_j}) \geq F(\bar{z}). \quad (21)$$

Choosing $k = k_j - 1$, from Eq. (9), we infer

$$\begin{aligned} \lim_{j \rightarrow \infty} \sup F_i(A_i^{k_j}) &\leq \lim_{j \rightarrow \infty} \sup (F_i(\bar{A}_i) + \frac{1}{2\sigma_i^k} \|\bar{A}_i - Y_i^k\|_F^2 \\ &\quad + \langle \nabla_{A_i} H(\{A_n^{k+1}\}_{n=1}^{i-1}, Y_i^k, \{A_n^k\}_{n=i+1}^N), \bar{A}_i - Y_i^k \rangle), \end{aligned} \quad (22)$$

where $\lim_{j \rightarrow \infty} A_i^{k_j} = \bar{A}_i$. Since $\lim_{j \rightarrow \infty} z^{k_j} = \bar{z}$ and $\lim_{j \rightarrow \infty} \|z^{k_j} - c^{k_j-1}\|_F = 0$ (Theorem 1(ii)), we can get that

$$\begin{aligned} \lim_{j \rightarrow \infty} \|\bar{A}_i - Y_i^k\|_F &\leq \lim_{j \rightarrow \infty} \|\bar{A}_i - A_i^{k_j}\|_F + \lim_{j \rightarrow \infty} \|A_i^{k_j} - Y_i^{k_j-1}\|_F \\ &= 0, \end{aligned} \quad (23)$$

From Eq. (22) and Eq. (23), we infer

$$\lim_{j \rightarrow \infty} \sup F_i(A_i^{k_j}) \leq \lim_{j \rightarrow \infty} \sup F_i(\bar{A}_i), \quad (24)$$

where $\lim_{j \rightarrow \infty} A_i^{k_j} = \bar{A}_i$. Therefore, from Eq. (20) and Eq. (24), we have

$$\lim_{j \rightarrow \infty} \sup F(z^{k_j}) \leq \lim_{j \rightarrow \infty} \sup F(\bar{z}). \quad (25)$$

From Eq. (20), Eq. (25) and Eq. (21), we infer

$$\lim_{j \rightarrow \infty} F(z^{k_j}) = F(\bar{z}). \quad (26)$$

Since Theorem 1, we have

$$\lim_{j \rightarrow \infty} H(z^{k_j}) = H(\bar{z}). \quad (27)$$

Thus from Eq. (26) and Eq. (27), we infer

$$\begin{aligned} \lim_{j \rightarrow \infty} H(z^{k_j}) + \lim_{j \rightarrow \infty} F(z^{k_j}) &= \lim_{j \rightarrow \infty} (H(z^{k_j}) + F(z^{k_j})) \\ &= \lim_{j \rightarrow \infty} J(z^{k_j}) \\ &= J(\bar{z}). \end{aligned}$$

This means J is constant on z^* .

From Eq. (17), we know that

$$f_i^{k+1} \in \partial_{A_i} J(\{A_j^{k+1}\}_{j=1}^i, \{A_j^k\}_{j=i+1}^N), \quad (28)$$

where the definitions of f_i^{k+1} is the same as in Eq. (15). From Theorem 1(ii) and Lemma 2, we infer

$$\begin{aligned} \lim_{k \rightarrow \infty} (\|\{f_i^{k+1}\}_{i=1}^N\|_F) &\leq \lim_{k \rightarrow \infty} \rho_b \|z^{k+1} - c^k\|_F \\ &= 0. \end{aligned} \quad (29)$$

Let

$$z_i^k = A_i^k. \quad (30)$$

From Theorem 1(i), we know that

$$\begin{aligned} \lim_{k \rightarrow \infty} J(\{z_i^{k+1}\}_{i=1}^{j-1}, z_j^{k+1}, \{z_i^k\}_{i=j+1}^N) &= \lim_{k \rightarrow \infty} J(z^k) \\ &= J(z^*). \end{aligned} \quad (31)$$

Therefore, from Eq. (28), Eq. (29), Eq. (30) and Eq. (31), we have

$$\begin{aligned} 0 &\in \partial_{z_j} \lim_{k \rightarrow \infty} J(\{z_i^{k+1}\}_{i=1}^{j-1}, z_j^{k+1}, \{z_i^k\}_{i=j+1}^N) \\ &= \partial_{z_j} J(z^*). \end{aligned}$$

This means $0 \in \partial J(z^*)$. □

9.4 Proof of Theorem 3

Proof. From Eq. (7), we know that $\sum_{q=1}^Q \frac{\alpha_q}{2} \text{Tr}((A_N)^T L_q A_N)$ is an infinitely differentiable function, and the norm of its derivatives of any order is also continuous, so $\sum_{q=1}^Q \frac{\alpha_q}{2} \text{Tr}((A_N)^T L_q A_N)$ is a real analytic function (real analytic functions are all KL functions [4]). Since the Frobenius norm, ℓ_p -norm is also all KL functions [3], it follows that Eq. (4) is a KL function. Therefore, from Definition 12, there exists a concave function ϕ so that

$$\phi'(J(z^k) - J(\bar{z})) \text{dist}(0, \partial J(z^k)) \geq 1. \quad (32)$$

From ϕ is the convex function, we have

$$\begin{aligned} \phi(J(z^{k+1}) - J(\bar{z})) &\leq \phi(J(z^k) - J(\bar{z})) \\ &\quad + \phi'(J(z^k) - J(\bar{z}))(J(z^{k+1}) - J(z^k)). \end{aligned} \quad (33)$$

From Lemma 2 and Theorem 2, we infer

$$\text{dist}(0, \partial J(z^k)) \leq \rho_b \|z^k - c^{k-1}\|_F. \quad (34)$$

Since the Eq. (32) and Eq. (34), we infer

$$\begin{aligned} \phi'(J(z^k) - J(\bar{z})) &\geq \frac{1}{\text{dist}(0, \partial J(z^k))} \\ &\geq \frac{1}{\rho_b \|z^k - c^{k-1}\|_F}. \end{aligned} \quad (35)$$

Let $J(k) = J(z^k) - J(\bar{z})$, from Eq. (33), Eq. (34), Eq. (35) and Theorem 1(i), we have

$$\begin{aligned} \phi(J(k)) - \phi(J(k+1)) &\geq \phi'(J(k))(J(k) - J(k+1)) \\ &\geq \frac{J(k) - J(k+1)}{\rho_b \|z^k - c^{k-1}\|_F} \\ &\geq \frac{\rho \|z^{k+1} - c^k\|_F^2}{\rho_b \|z^k - c^{k-1}\|_F}. \end{aligned}$$

Define $C = \frac{\rho}{\rho_b}$, C is a constant, so we infer

$$\|z^{k+1} - c^k\|_F^2 \leq C(\phi(J(k)) - \phi(J(k+1))) \|z^k - c^{k-1}\|_F.$$

Using the fact that $2ab \leq a^2 + b^2$

$$2\|z^{k+1} - c^k\|_F \leq C(\phi(J(k)) - \phi(J(k+1))) + \|z^k - c^{k-1}\|_F.$$

Sum both sides

$$\begin{aligned} 2 \sum_{k=l+1}^K \|z^{k+1} - c^k\|_F &\leq \sum_{k=l+1}^K \|z^k - c^{k-1}\|_F \\ &\quad + C(\phi(J(l+1)) - \phi(J(K+1))) \\ &= C(\phi(J(l+1)) - \phi(J(K+1))) \\ &\quad + \|z^{l+1} - c^l\|_F + \sum_{k=l+1}^K \|z^{k+1} - c^k\|_F. \end{aligned} \quad (36)$$

From Eq. (36), we can get that

$$\begin{aligned} \lim_{K \rightarrow \infty} \sum_{k=l+1}^K \|z^{k+1} - c^k\|_F &\leq \|z^{l+1} - c^l\|_F + C\phi(J(l+1)) \\ &\quad - \lim_{K \rightarrow \infty} C\phi(J(K+1)) \\ &< \infty. \end{aligned}$$

Thus we have

$$\sum_{k=l+1}^{\infty} \|z^{k+1} - c^k\|_F < \infty. \quad (37)$$

From Algorithm 1, no matter $c^k = z^k + \beta_k(z^k - z^{k-1})$ or $c^k = z^k$, we always have

$$\|z^{k+1} - z^k\|_F - \beta_k \|z^k - z^{k-1}\|_F \leq \|z^{k+1} - c^k\|_F. \quad (38)$$

From Eq. (37) and Eq. (38), we know that

$$\begin{aligned} \sum_{k=l+1}^{\infty} (\|z^{k+1} - z^k\|_F - \beta_k \|z^k - z^{k-1}\|_F) &\leq \sum_{k=l+1}^{\infty} \|z^{k+1} - c^k\|_F \\ &< \infty. \end{aligned}$$

Since

$$\|z^{k+1} - z^k\|_F - \beta_{max} \|z^k - z^{k-1}\|_F \leq \|z^{k+1} - z^k\|_F - \beta_k \|z^k - z^{k-1}\|_F,$$

we also have

$$\begin{aligned} &\sum_{k=l+1}^{\infty} (\|z^{k+1} - z^k\|_F - \beta_{max} \|z^k - z^{k-1}\|_F) \\ &\leq \sum_{k=l+1}^{\infty} (\|z^{k+1} - z^k\|_F - \beta_k \|z^k - z^{k-1}\|_F) \\ &< \infty, \end{aligned} \quad (39)$$

and

$$\begin{aligned} &\sum_{k=l+1}^{\infty} \|z^{k+1} - z^k\|_F - \sum_{k=l+1}^{\infty} \beta_{max} \|z^k - z^{k-1}\|_F \\ &= \sum_{k=l+1}^{\infty} (\|z^{k+1} - z^k\|_F - \beta_{max} \|z^{k+1} - z^k\|_F) \\ &\quad + \beta_{max} \|z^{l+1} - z^l\|_F \\ &= \sum_{k=l+1}^{\infty} (1 - \beta_{max}) \|z^{k+1} - z^k\|_F + \beta_{max} \|z^{l+1} - z^l\|_F. \end{aligned} \quad (40)$$

Let $s^k = \|z^{k+1} - z^k\|_F$, from Eq. (39) and Eq. (40), we have

$$\begin{aligned} \sum_{k=l+1}^{\infty} (1 - \beta_{max}) s^k - \beta_{max} s^l &= \sum_{k=l}^{\infty} (s^{k+1} - \beta_{max} s^k) \\ &\leq \sum_{k=l+1}^{\infty} \|z^{k+1} - c^k\|_F \\ &< \infty. \end{aligned}$$

Thus we infer

$$\sum_{k=l+1}^{\infty} (1 - \beta_{max}) s^k < \infty. \quad (41)$$

From $(1 - \beta_{max})$ is a positive constant and Eq. (41), we infer

$$\sum_{k=l+1}^{\infty} \|z^{k+1} - z^k\|_F = \sum_{k=l+1}^{\infty} s^k < \infty.$$

Thus we have

$$\sum_{k=0}^{\infty} \|z^{k+1} - z^k\|_F = \sum_{k=0}^{\infty} s^k < \infty.$$

This shows that

$$\begin{aligned} \lim_{K \rightarrow \infty} \|z^{K+p} - z^K\|_F &= \lim_{K \rightarrow \infty} \left\| \sum_{k=K+1}^{K+p} (z^{k+1} - z^k) \right\|_F \\ &\leq \lim_{K \rightarrow \infty} \sum_{k=K+1}^{K+p} \|z^{k+1} - z^k\|_F \\ &= \lim_{K \rightarrow \infty} \sum_{k=K+1}^{\infty} s^k \\ &= 0. \end{aligned} \quad (42)$$

This means the Theorem 3 is true. $\square$

9.5 Proof of Theorem 4

Proof. We use the MTTKRP technology [22] to calculate $X^{(n)} B_n$. Therefore, the computational complexity of $X^{(n)} B_n$ reaches

$$R \left(\prod_{i=1}^N I_i \right). \quad (43)$$

Similarly, $(B_n)^T B_n$ can be calculated efficiently by $((A_N)^T A_N) * \dots * ((A_{n+1})^T A_{n+1}) * ((A_{n-1})^T A_{n-1}) * \dots * ((A_1)^T A_1)$, where $*$ represents the Hadamard product that is the elementwise matrix product. Therefore, the computational complexity of $(B_n)^T B_n$ is

$$R^2 \sum_{j=1, j \neq n}^N I_i. \quad (44)$$

Therefore, from Eq. (43) and Eq. (44), we know that the computational cost of $\nabla_{A_i} H(A_{(N)})$ is

$$\mathcal{O}(R(\prod_{i=1}^N I_i) + R^2 \sum_{j=1, j \neq i}^N I_i). \quad (45)$$

The cost of computing the Lipschitz constant, projection to nonnegative, and tensor unfolding is negligible compared to the cost of computing partial gradient $\nabla_{A_i} H(A_{(N)})$. Similarly, the cost of the sparsity projection is also negligible compared to the cost of computing the $\nabla_{A_i} H(A_{(N)})$, as the time complexity of finding the largest s elements in an array of p elements is $\mathcal{O}(p + s \log_2(s))$. Therefore, the time complexity of the Algorithm 1 in each iteration is approximately estimated as

$$\mathcal{O}(NR(\prod_{i=1}^N I_i) + R^2 \sum_{i=1}^N (\sum_{j=1, j \neq i}^N I_i)).$$

□

Table 4: Average and standard deviation results by all methods on the remaining datasets. The best performance is highlighted in bold.

Dataset Method	ConcreteCrackImages			PNW		
	ACC	ARI	NMI	ACC	ARI	NMI
MGNMF	0.781 ± 0.012	0.302 ± 0.015	0.295 ± 0.013	0.432 ± 0.018	0.113 ± 0.021	0.299 ± 0.038
GSNMF	0.669 ± 0.024	0.192 ± 0.021	0.180 ± 0.014	0.421 ± 0.011	0.104 ± 0.011	0.280 ± 0.017
ℓ_1 -SNCP	0.759 ± 0.037	0.282 ± 0.041	0.270 ± 0.019	0.367 ± 0.032	0.047 ± 0.035	0.217 ± 0.046
SGCP	0.774 ± 0.023	0.298 ± 0.013	0.283 ± 0.017	0.442 ± 0.023	0.109 ± 0.019	0.284 ± 0.015
GSNTD	0.759 ± 0.001	0.286 ± 0.002	0.277 ± 0.001	0.467 ± 0.013	0.119 ± 0.006	0.351 ± 0.010
ℓ_0 -SGNCP	0.787 ± 0.008	0.308 ± 0.006	0.291 ± 0.006	0.448 ± 0.015	0.128 ± 0.014	0.330 ± 0.026
MGNCP	0.805 ± 0.005	0.320 ± 0.006	0.322 ± 0.008	0.443 ± 0.021	0.133 ± 0.013	0.361 ± 0.015

Dataset Method	ORL			USPS		
	ACC	ARI	NMI	ACC	ARI	NMI
MGNMF	0.389 ± 0.025	0.246 ± 0.026	0.641 ± 0.008	0.456 ± 0.012	0.271 ± 0.016	0.403 ± 0.015
GSNMF	0.216 ± 0.003	0.045 ± 0.001	0.456 ± 0.002	0.636 ± 0.014	0.452 ± 0.016	0.561 ± 0.013
ℓ_1 -SNCP	0.503 ± 0.012	0.346 ± 0.018	0.711 ± 0.008	0.627 ± 0.033	0.442 ± 0.033	0.553 ± 0.025
SGCP	0.511 ± 0.014	0.360 ± 0.013	0.723 ± 0.007	0.618 ± 0.032	0.425 ± 0.026	0.543 ± 0.018
GSNTD	0.502 ± 0.003	0.340 ± 0.001	0.713 ± 0.002	0.580 ± 0.005	0.401 ± 0.008	0.508 ± 0.006
ℓ_0 -SGNCP	0.520 ± 0.015	0.363 ± 0.017	0.723 ± 0.007	0.640 ± 0.017	0.453 ± 0.013	0.561 ± 0.008
MGNCP	0.521 ± 0.008	0.366 ± 0.009	0.731 ± 0.005	0.647 ± 0.008	0.458 ± 0.007	0.568 ± 0.002

10 Supplementary Experimental Results

In this section, we continue to complete the remaining experiments outlined in Section 6.

10.1 Supplementary experiment 1: Numerical Assessment of Clustering Efficacy

We continue to evaluate the clustering performance of all methods on the remaining four real-world datasets described in Section 6, and all parameter settings are the same as in Section 6. Table 4 shows the clustering performance results (average and standard deviation) for the methods over 10 independent runs with different initial points on the remaining datasets. From Table 4, we can get the same conclusion as Section 6 that MGNCP remarkably outperforms all other methods on all datasets. This once again fully demonstrates the effectiveness of MGNCP and shows that incorporating our proposed multi-graph regularization into NCP can indeed enhance the feature extraction capability and interpretability.

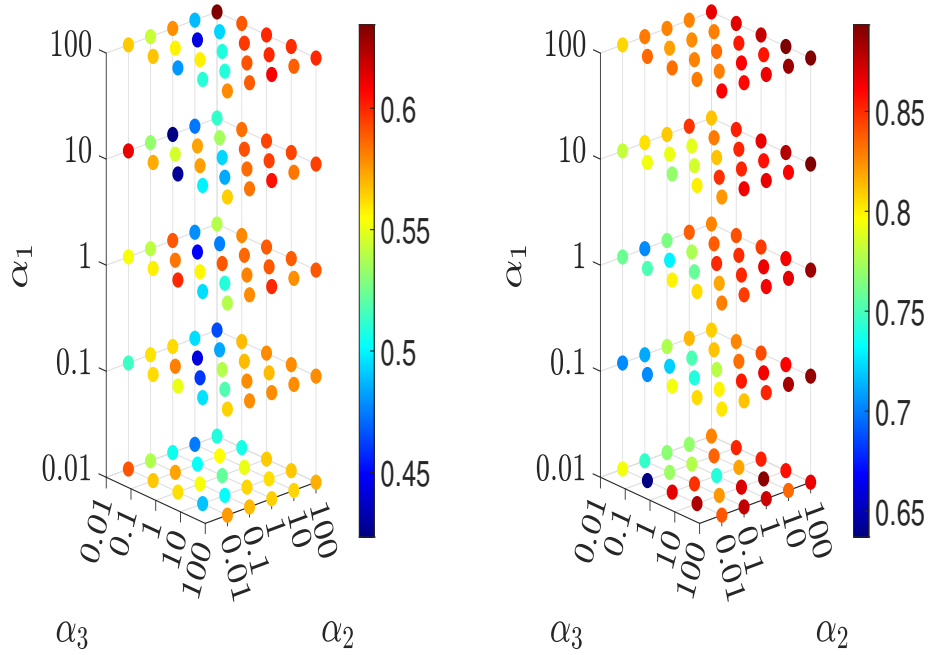


Fig. 1: Clustering NMI with respect to the parameters α_q on the UMIST (left) and Pixraw10P (right) datasets, respectively. The color hue of the point indicates the NMI value of the corresponding entry, and the mapping relationship between the NMI value and the color hue is displayed in the colorbar above.

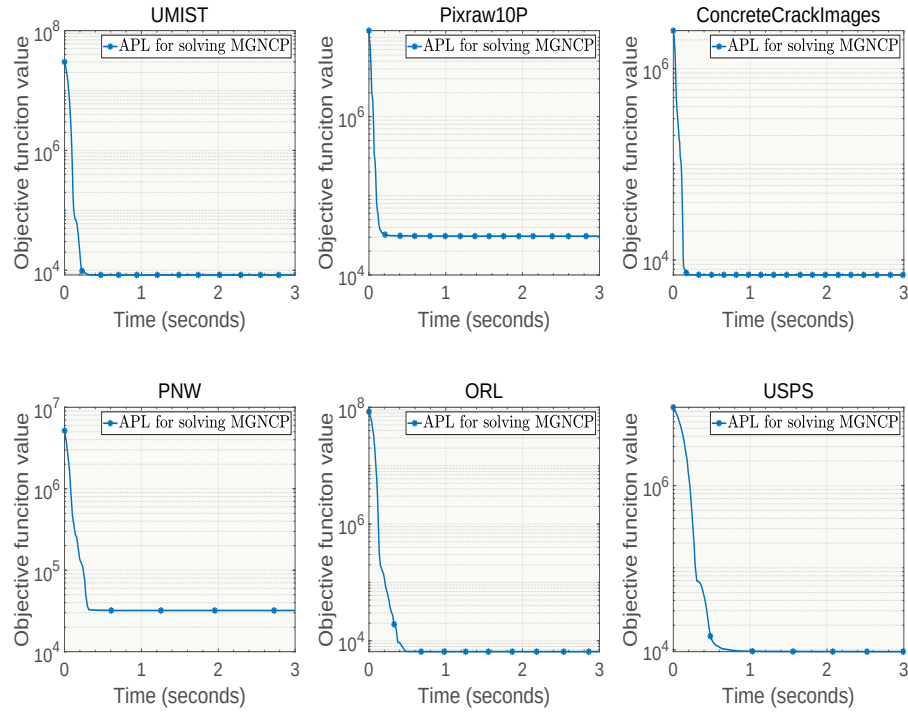


Fig. 2: Convergence curves of the APL algorithm for solving MGNCP on six datasets.

10.2 Supplementary experiment 2: Parameter sensitivity analysis

Next, we analyze the impact of the graph regularization parameters α_q in MGNCP on clustering performance. Specifically, we alternately vary one of the graph regularization parameters α_q , and record the clustering performance under all possible parameter combinations, as shown in Figure 1. For the sake of space, we only illustrate results on the UMIST and Pixraw10P datasets. From Figure 1, we have the following observations.

- 1) The clustering performance of MGNCP on the UMIST dataset gradually becomes relatively stable as both α_2 and α_3 increase. Similarly, when α_2 and α_3 are large enough, the clustering performance on the UMIST dataset gradually increases and becomes relatively stable as α_1 increases. However, when these parameters are small, the clustering performance on the UMIST dataset is less stable.
- 2) The clustering performance of MGNCP on the Pixraw10P dataset gradually becomes relatively stable as α_1 increases. Besides, when $\alpha_1 \geq 0.1$, the clustering performance on the Pixraw10P dataset shows a positive correlation with both α_2 and α_3 . However, when α_1 is small, the clustering performance on the Pixraw10P dataset is relatively sensitive to α_2 and α_3 .

10.3 Supplementary experiment 3: Convergence analysis

We also evaluate the convergence performance of our proposed APL algorithm (Algorithm 1) for solving the MGNCP. Figure 2 plots how the objective function value changes with respect to time for the APL algorithm on several datasets. From Figure 2, we observe that the objective function value monotonically decreases to a stationary point, thereby validating the effectiveness and convergence of the APL algorithm for solving the MGNCP.

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